Formula 1 Podium Prediction

Analytical Report · ACL Milestone 1 · CSEN 903 Advanced Computer Lab, German University in Cairo

Task. For every driver who starts a Grand Prix, predict whether they finish on the podium (positionOrder ≤ 3) using only information that is known **after qualifying and before the race starts**. The unit of analysis is one driver in one race; the pair (raceId, driverId) is unique. **Data.** The 14 relational tables of the Formula 1 World Championship dataset (Ergast / Kaggle, 1950–2024). **This report** accompanies the Kaggle notebook; every figure and number below is taken from that notebook's saved run.

> **Important – association, not causation.** Everything in this report (feature importance, SHAP, LIME, ablations, the home-race analysis) describes what the *model* relies on or what is *observed in historical data*. None of it shows that a feature *causes* podium finishes. For example, a high team points share does not make a driver faster; it is a sign of a fast car.

Results at a glance

Item	Result
Modeling table	26,668 driver-race rows, exactly one per (raceId, driverId); 13 numeric, leakage-safe features in 5 groups
Split (chronological)	Train 1950–2019 (24,529 rows, 12.5% podiums) · Validation 2020–2021 (780 rows, 15.0%) · Test 2022–2024 (1,359 rows, 15.0%)
Selected model	Shallow neural network (Adam), chosen by **validation** PR-AUC (0.715); F1 threshold 0.34, also tuned on validation only
Unseen test result	PR-AUC **0.700** · ROC-AUC **0.932** · F1 **0.645** (precision 0.605, recall 0.691, accuracy 0.886)
Simple baseline (grid ≤ 3)	Test PR-AUC 0.647 · ROC-AUC 0.914 · F1 0.649 – a strong baseline: the starting grid carries most of the signal
Q1 front row → podium	Conversion rose from 55.7% (1950–2013) to 76.0% (2014–2024); circuit ranking does **not** carry over (rank correlation −0.03)
Q2 home advantage	Adjusted odds ratio 0.917 (95% CI 0.780–1.087): no evidence of a home-race advantage once grid position is controlled
Q3 mechanical retirements	Fell from 9.1% (2014–2021) to 5.6% (2022–2024); highest: Caterham, Lotus, Toro Rosso, then Alfa Romeo and Alpine

Table 1. *Headline results (all from the saved Kaggle run).*

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- 4 Split and preprocessing · 5 Models and results · 6 Feature-group ablation · 7 Explainability (XAI)
- 8 Inference function · 9 Known limitations · 10 Where each brief requirement is covered

1 Data audit and cleaning

1.1 What the raw data looked like

The CSV files write a missing value as the literal text **\N**. pandas does not recognise it, so isnull() reports almost nothing while more than 100,000 cells of results.csv are actually missing. Because of the sentinel, numeric columns (grid, position, milliseconds ...) were also read as text. The EDA below is computed with code on all 14 tables, not by manual inspection.

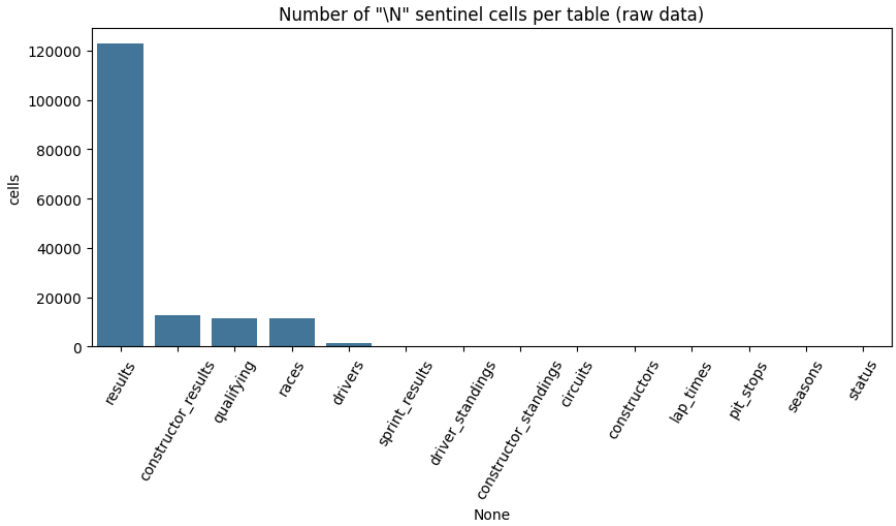


Figure 1. *Number of "\N" sentinel cells per raw table. results.csv is by far the most affected.*

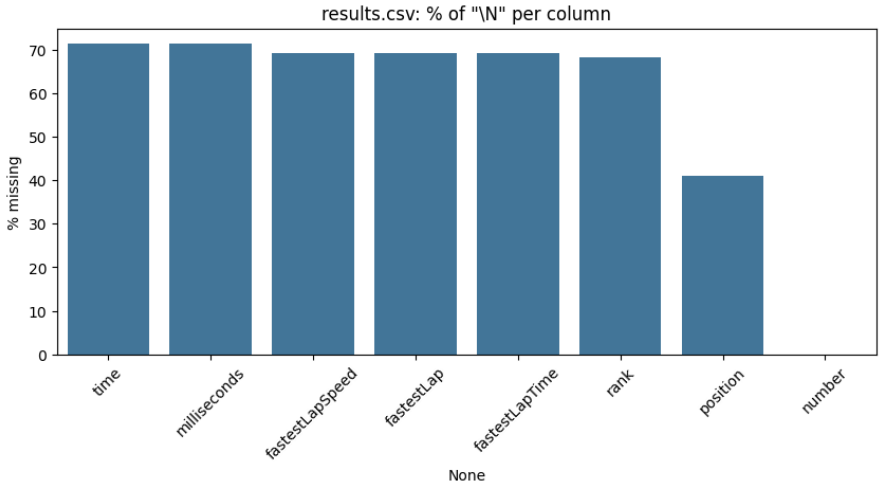


Figure 2. *Share of "\N" per column in results.csv. Columns such as fastestLap, rank and time are missing for most rows; position is missing for retired drivers.*

1.2 Cleaning steps

- **Sentinels:** every \N becomes NaN. After cleaning, **0 sentinels remain** (checked by code).
- **Semantic typing:** id and numeric columns become numbers; races.date and drivers.dob become datetime objects; lap-time strings become seconds (example: 1:26.572 → 86.572); stray spaces are stripped from text (e.g. a nationality with a trailing space).
- **Study period:** seasons 1950–2024 only, and the filter is applied to every race-keyed table.
- **Raw data stays immutable:** the original tables are kept in a dictionary and never modified; all cleaning is done on copies.

1.3 Primary keys and foreign keys

Check	Result
Primary keys, 14 tables (circuits, constructors, drivers, races, status, seasons, results, qualifying, standings, constructor_results, sprint_results, pit_stops, lap_times)	**0 null keys and 0 duplicate keys** in every table
Foreign keys, 23 parent–child checks (e.g. results.driverId → drivers, results.statusId → status, races.circuitId → circuits)	**0 orphan rows** in every check
Chronology trap	raceId is **not** in time order (the 2009 races have ids 1–17; 2008 starts at 18). All time ordering therefore uses the race date, never raceId.

Table 2. *Key audit.*

1.4 Grain: exactly one row per driver per race

In the 1950s a driver could take over a teammate's car mid-race. The data then holds the same driver twice in one race. The check finds **85 duplicated (raceId, driverId) pairs**, all in early seasons (Figure 3). Example: at the 1957 British Grand Prix two drivers appear twice each, once in each of the two cars they shared.

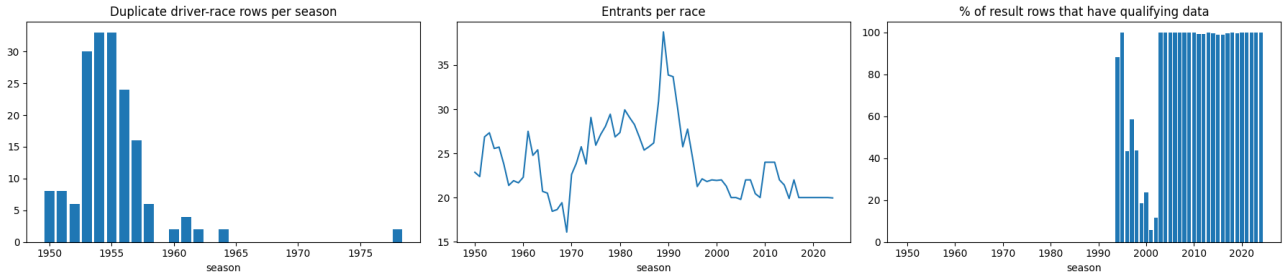


Figure 3. *Left: duplicate driver-race rows per season (almost all in the 1950s). Middle: entrants per race (large, uneven fields in the 1980s). Right: share of results rows that have qualifying data – none before 1994 and full coverage only from 2003.*

- **Rule chosen:** keep the row with the **lowest resultId** for each driver and race.
- **Why:** the rule does not look at the outcome. The alternative, "keep the best finish", selects rows by the target itself; it would create **16 extra podium labels** (3,395 podiums vs 3,379 with our rule) and could pair a grid slot with another car's result.
- **Honest limit:** the lowest resultId is a consistent, outcome-blind rule, but it is not guaranteed to identify the driver's original car in every case.
- **Result:** 26,759 result rows → **26,668 modeling rows**, with a code assertion that (raceId, driverId) is unique.

1.5 Who is in the modeling table

The brief defines the unit as “one driver *entering* one race”. We therefore keep every entry in results.csv. Dropping rows because their final status says “Did not start” or “Withdrew”, or because they completed zero laps, would let **post-race information decide who is in the data**, which is a form of leakage. The consequences are disclosed: 405 Indianapolis 500 result rows (a separate American race that counted for the championship in 1950–1960) and 1,638 rows with grid = 0 are retained, as are 23 entries whose status is “Withdrew” but who completed laps. The grid = 0 indicator is treated as a *proxy*: it mixes historical non-starters with real pit-lane starts. For the descriptive questions Q2 and Q3 (history, not prediction) the analysis is restricted to actual starters (positive grid or at least one lap).

1.6 EDA after cleaning

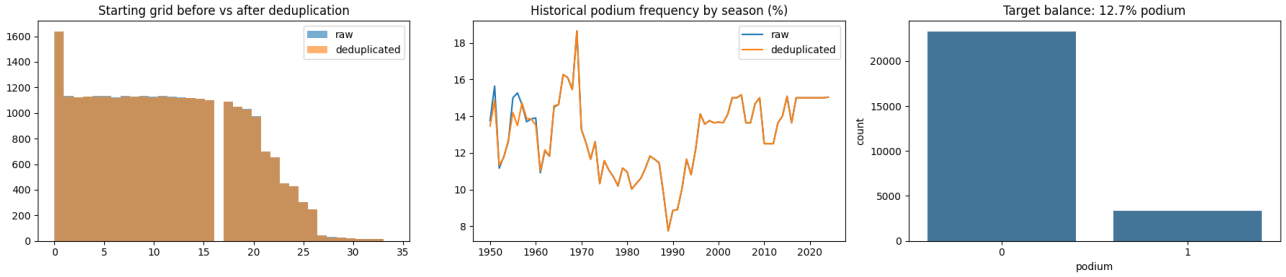


Figure 4. *Before vs after: starting-grid distribution (raw vs deduplicated), podium frequency per season, and target balance (12.7% podiums overall).*

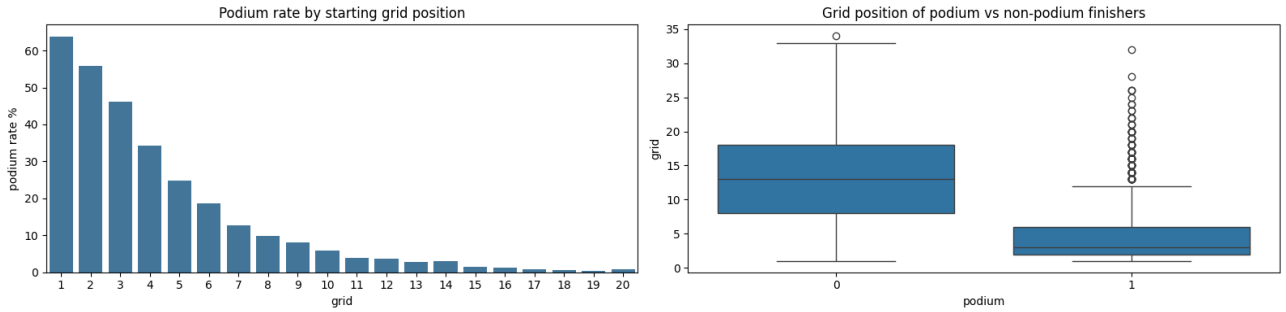


Figure 5. *Podium rate by starting grid position (left) and grid position of podium vs non-podium finishers (right). Starting near the front is by far the strongest pattern.*

Era	Driver-races	Podium rate	Qualifying data available
1950–1969	4,030	13.8%	0.0%
1970–1989	8,309	10.8%	0.0%
1990–2002	5,359	11.9%	28.9%
2003–2013	4,344	13.8%	99.8%
2014–2024	4,626	14.8%	99.7%

Table 3. *Historical eras: the podium base rate is stable at roughly 11–15%, but qualifying data only exists from the 1990s and is complete from 2003 (qualifying-coverage drift).*

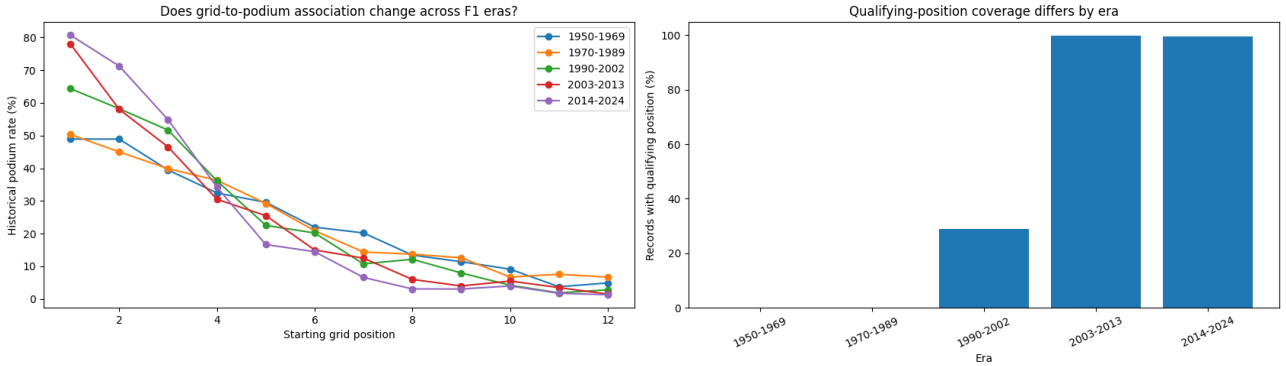


Figure 6. *The grid-to-podium relationship across rule eras (left) and qualifying coverage by era (right). Coverage drift is the main reason qualifying features need a fallback.*

2 The three data-engineering questions

Each question needs several tables joined together (results alone cannot answer them). For each we state the joins, the grain, how missing data was handled, and whether the pre-race cutoff matters. The questions are **descriptive and retrospective**: they use race outcomes to describe history and are never used as model features.

Q1 Which circuits convert a front-row start into a podium most often, and does the ranking change after 2014?

- **Joins:** results → races (year, circuit) → circuits (name); and results → qualifying (qualifying position).
- **Grain:** one driver start from grid slot 1 or 2 (front row). The conversion rate of a circuit is podiums ÷ front-row starts. Circuits need at least 10 front-row starts in an era to be ranked (1,796 front-row starts in 1950–2013, 454 in 2014–2024). Eras: 1950–2013 vs 2014–2024.
- **Grid vs qualifying:** "grid" is where the car actually started (after penalties); "qualifying position" is where the driver qualified. They are compared on 2003–2024, where qualifying data is essentially complete (8,946 driver-races).
- **Missingness and cutoff:** grid is present for every row; qualifying position exists only from 1994 (complete from 2003), so it is not used before 2003. Grid and qualifying are both known before the start; the podium result is only what is being counted.

Rank	1950–2013	Conversion	2014–2024	Conversion
1	Circuit de Nevers Magny-Cours	81% (n=36)	Suzuka Circuit	89% (n=18)
2	Istanbul Park	79% (n=14)	Yas Marina Circuit	86% (n=22)
3	Dijon-Prenois	75% (n=12)	Circuit de Monaco	85% (n=20)
4	Circuit Bremgarten	70% (n=10)	Circuit de Spa-Francorchamps	82% (n=22)
5	Aintree	70% (n=10)	Circuit de Barcelona-Catalunya	82% (n=22)

Table 4. *Top five circuits per era (at least 10 front-row starts).*

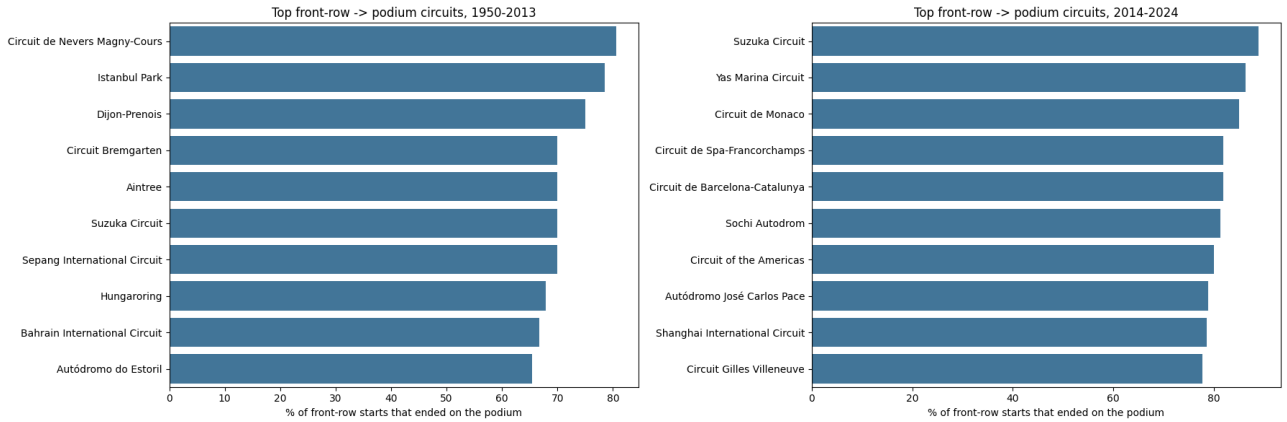


Figure 7. *Top ten circuits by front-row-to-podium conversion in each era.*

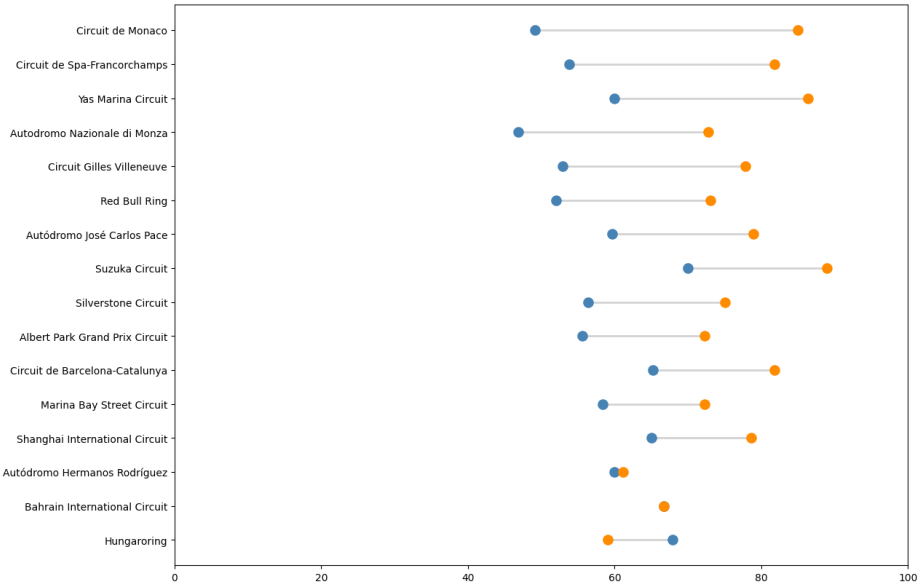


Figure 8. *The 16 circuits ranked in both eras: blue = 1950–2013, orange = 2014–2024. Positions are scattered, so an old rank says little about the new rank.*

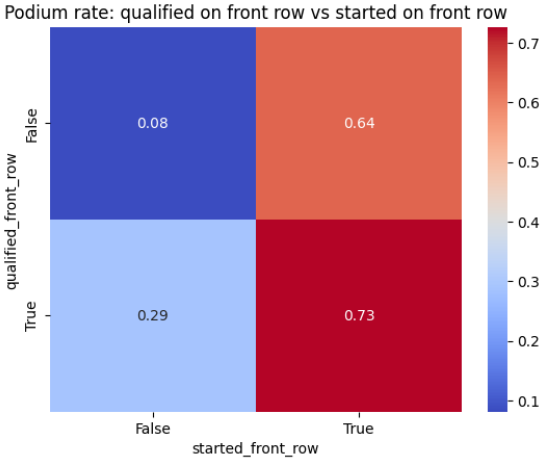


Figure 9. *Podium rate by qualifying front row (rows) and grid front row (columns), 2003–2024. A driver who qualified on the front row but started elsewhere (a penalty) podiums far less.*

Qualified P1–2	Started P1–2	Driver-races	Podium rate
yes	yes	818	0.73
yes	no (penalty / other)	38	0.29
no	yes (promoted)	36	0.64
no	no	8,054	0.08

Table 5. *Qualifying position vs grid position (2003–2024).*

Answer Q1. Front-row starts converted into podiums **55.7%** of the time before 2014 and **76.0%** from 2014. Before 2014 the leaders were Magny-Cours, Istanbul Park and Dijon-Prenois; from 2014 they are Suzuka, Yas Marina, Monaco and Spa. **The ranking changes**: across the 16 circuits ranked in both eras the Spearman rank correlation is **−0.03** (no carry-over). Qualifying and grid are not interchangeable: only 38 of 856 front-row qualifiers started elsewhere, but they podiumed 29% of the time vs 73% for those who started there. *Caveat:* modern circuits have at most about 22 front-row starts, so neighbouring ranks are not statistically distinguishable, and the era change reflects reliability and team dominance as much as the circuit.

Q2 Do drivers podium more often at home, after controlling for starting position?

- **Joins:** results → races → circuits.country (country of the race) and results → drivers.nationality, mapped to a country with a hand-built dictionary (e.g. British → UK); 0 unmapped nationalities.
- **Grain:** one actual start (positive grid or at least one lap): 25,105 starts, of which 2,324 (9.3%) are home starts.
- **Control:** comparing like with like. Podium rates are compared inside grid bands (P1, P2–3, P4–6, P7–10, P11+). A logistic regression of podium on home, grid band and decade then gives an adjusted **odds ratio**; its 95% interval comes from 200 bootstrap resamples.
- **Cutoff:** nationality, race country and grid are all known before the start.

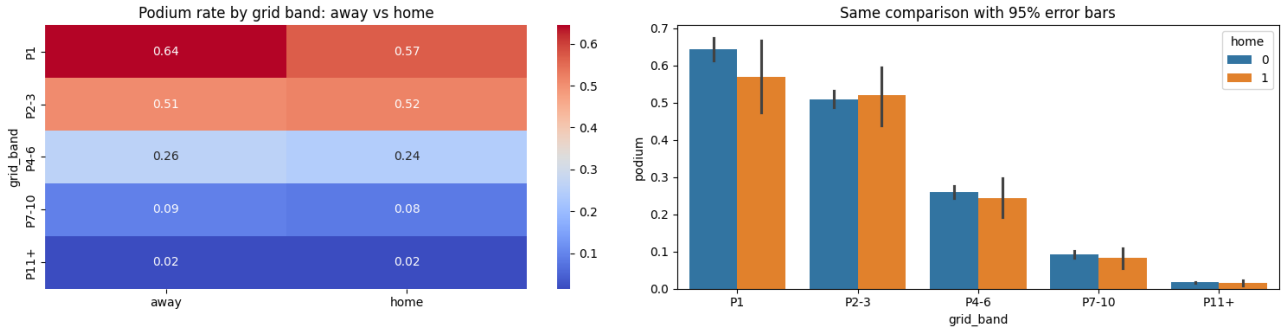


Figure 10. *Left: podium rate by grid band, away vs home. Right: the same with 95% error bars. Within each band the two are almost identical.*

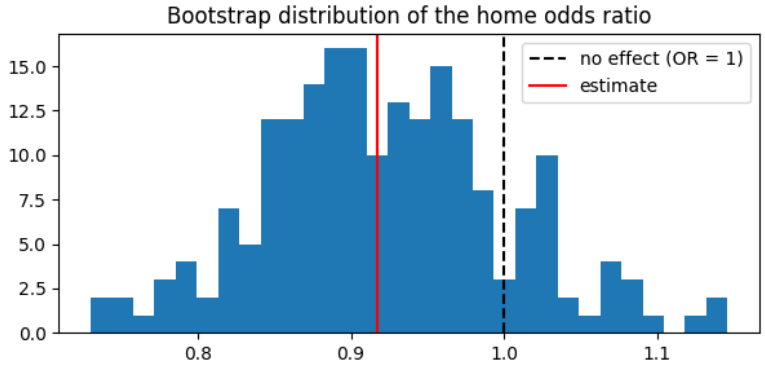


Figure 11. *Bootstrap distribution of the adjusted home odds ratio. The dashed line (OR = 1) is inside the distribution.*

Answer Q2. No evidence of a home advantage. Unadjusted, home drivers podium *less* often (11.2% vs 13.7%), largely because many home starts are one-off local entries in slow cars. After controlling for grid band and decade the odds ratio is **0.917 (95% CI 0.780–1.087)**; the interval contains 1, so the data do not support an adjusted home effect in either direction. *Caveats:* nationality is a single label; home starts are a minority; the result is an association.

Q3 Which constructors have the highest mechanical-retirement rate, 2014–2021 vs 2022–2024?

- **Joins:** results → races (year) → status (reason text) → constructors (team name).
- **Definition:** a mechanical retirement is a start that ended because a car component failed. We use an explicit list of 37 status labels (Engine, Gearbox, Power Unit, Brakes, Hydraulics, Electrical, Suspension, Overheating, ...). **Excluded** by the brief's definition: Collision (150 cases), Accident (85), Collision damage (59), Spun off (11), Puncture (7), Disqualified (11), Withdrew (11). **Ambiguous** labels such as Retired (50), Wheel (11) and Tyre (4) are not counted, so no unclear case inflates the rate.
- **Grain and denominator:** one actual constructor start (positive grid or at least one lap). Teams need at least 30 starts in an era.
- **Cutoff:** not applicable. This is a post-race description; the status column is never a model input.

Rank	2014–2021	Rate (failures/starts)	2022–2024	Rate (failures/starts)
1	Caterham	23.5% (8/34)	Alfa Romeo	12.5% (11/88)
2	Lotus F1	17.1% (13/76)	Alpine F1 Team	11.0% (15/136)
3	Toro Rosso	16.1% (39/242)	AlphaTauri	6.8% (6/88)
4	McLaren	13.8% (44/320)	Ferrari	5.2% (7/135)
5	Renault	13.5% (27/200)	Williams	5.2% (7/135)

Table 6. *Five highest mechanical-retirement rates per era (teams with at least 30 starts).*

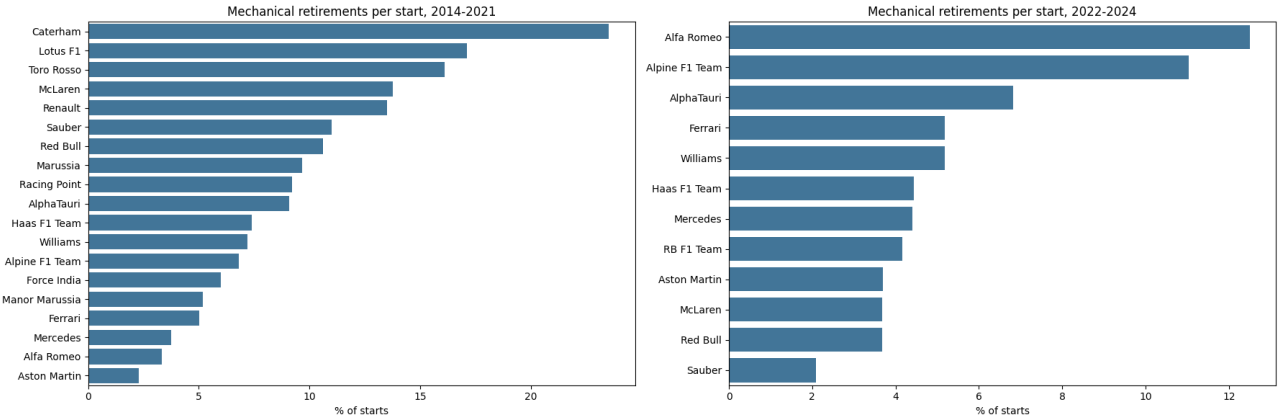


Figure 12. *Mechanical retirements per start for every constructor with at least 30 starts: 2014–2021 (left) and 2022–2024 (right).*

Answer Q3. Across all teams the mechanical-retirement rate fell from **9.1%** (2014–2021) to **5.6%** (2022–2024). The least reliable teams in 2014–2021 were **Caterham, Lotus F1 and Toro Rosso**, then McLaren (the troubled Honda years) and Renault. In 2022–2024 they were **Alfa Romeo (12.5%) and Alpine (11.0%)**, then AlphaTauri. The lists differ between eras: McLaren drops out of the top five. *Caveats:* three seasons is a short sample; Caterham has only 34 starts (8 failures); renamed teams (Toro Rosso → AlphaTauri → RB, Lotus F1 → Renault → Alpine, Sauber → Alfa Romeo) have separate constructor ids; some status labels are vague.

3 Features and leakage control

Every feature must be known **before the race starts**. Thirteen numeric features in five groups are used; because all are numeric, no categorical encoding is needed.

Group	Features	Why it is valid at the pre-race cutoff
grid	grid_pos, pit_lane_start	The starting grid is fixed before the lights go out. grid = 0 is placed at the back of the field and flagged separately.
qualifying	quali_position, quali_gap_pct (Q1 time vs fastest Q1, %)	Qualifying is finished before the race. A missing qualifying position falls back to the grid position.
driver_form	drv_podium_rate_last5, drv_finish_pct_last5, drv_starts_before	Computed with shift(1): each row sees only the driver's *previous* races, never the current one.
constructor_form	cons_podiums_last5, cons_best_pct_last5	Aggregated to one row per (team, race) *first*, then shifted, so a teammate's result in the same race cannot leak.
standings	drv_champ_rank_before, drv_champ_pts_share, cons_champ_rank_before, cons_champ_pts_share	Taken from the standings after the *previous round of the same season*. Round 1 means 0 points. Points are expressed as a share of the leader's points.

Table 7. *Selected features and why each is valid before the race.*

Excluded as future information: positionOrder, points, laps, status, fastest lap, pit stops, and post-race standings. Sprint results were deliberately not used.

3.1 Leakage controls (all verified by code)

- **Past-only form:** shift(1) before every rolling window. Check: Lewis Hamilton's 2008 form agrees with a manual "previous five races" calculation (True), his first career race has no form value (True), and all round-1 points shares are 0 (True).
- **Chronological order:** a race index built from the race date, because raceId is not chronological.
- **Comparable across eras:** points systems changed (9–10 points for a win before 2010, 25 after), so we use the share of the leader's points and finish percentiles instead of raw points and positions.
- **Final check in the notebook:** unique driver-race grain, time order, disjoint splits and no outcome columns as inputs all pass.

3.2 Evidence for the feature choices

Features were chosen from the EDA hypothesis that the starting order, the strength of the car (constructor form and standings) and the driver's recent results all matter. Figures 13–15 show the evidence on the training seasons only: qualifying position (−0.48) and grid position (−0.41) are the strongest single signals, followed by the driver's recent podium rate and points share (+0.46) and the team's points share (+0.42). The binned plots show podium rates rising steadily with each form feature. The heatmap also shows strong redundancy (grid_pos and quali_position correlate at 0.97), which is why the groups are later tested together in the ablation.

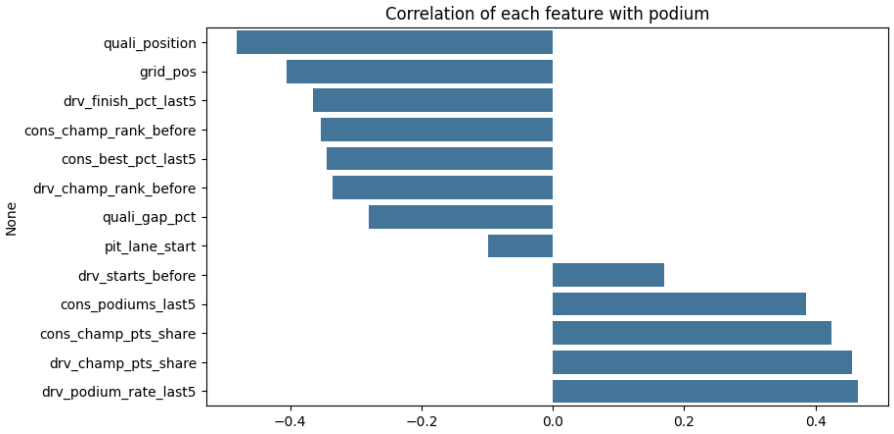


Figure 13. *Correlation of each feature with the podium target (training seasons).*

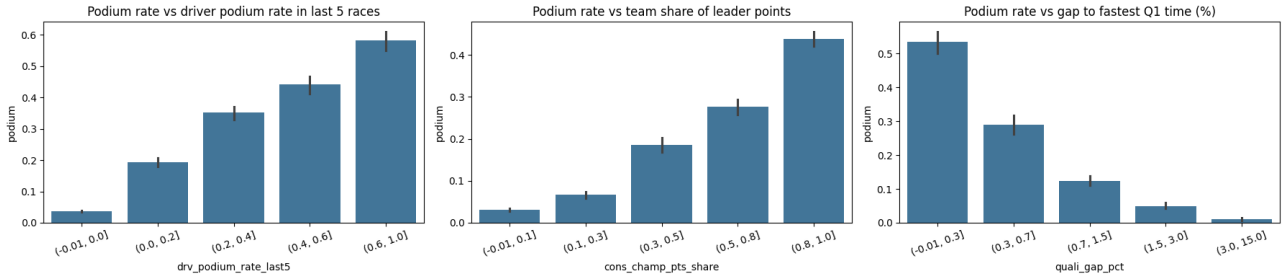


Figure 14. *Podium rate against driver form (left), team points share (middle) and qualifying gap to the fastest Q1 time (right).*

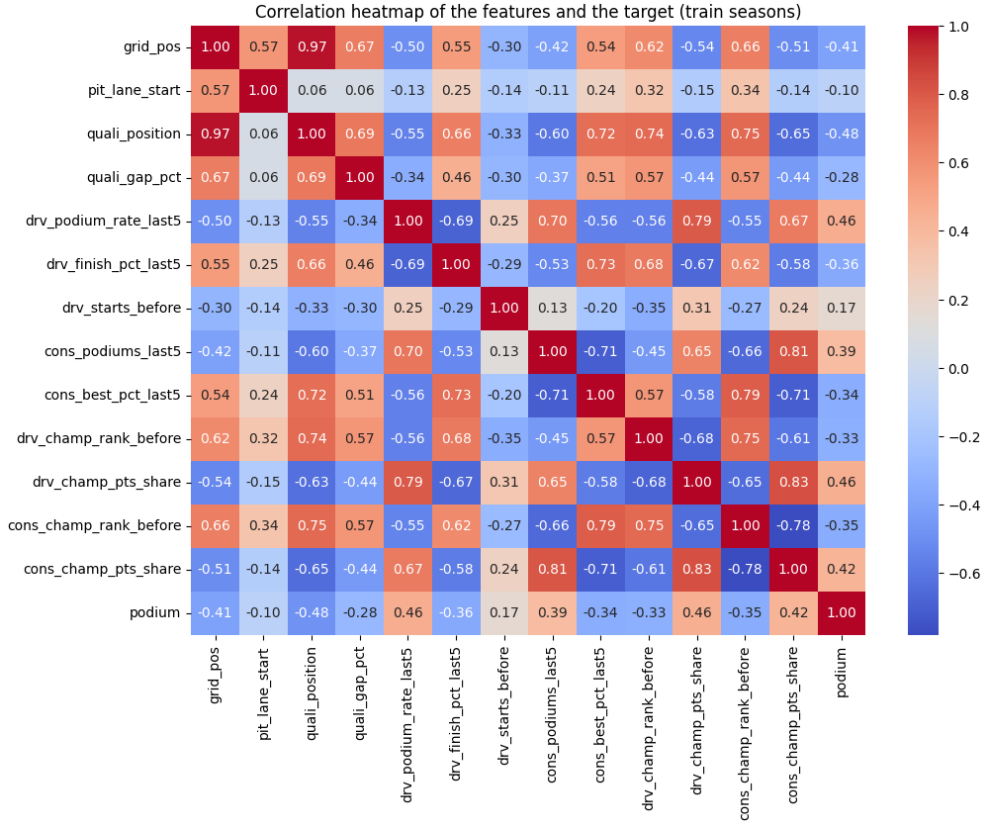


Figure 15. *Correlation heatmap of the 13 features and the target (training seasons).*

Feature	Training rows with a value
quali_gap_pct / quali_position	33.5% / 34.1%
driver and team championship rank before	85.5% / 86.0%
points shares	92.0% – 92.1%
driver form (last 5)	96.5%
constructor form (last 5)	98.6%
grid_pos, pit_lane_start, drv_starts_before	100%

Table 8. *Feature coverage in the training seasons. Qualifying features exist for only about a third of training rows: this is the qualifying-coverage drift.*

4 Train / validation / test split and preprocessing

Split	Seasons	Rows	Podium rate	Used for
Train	1950–2019	24,529	12.5%	Fitting models, medians and the scaler
Validation	2020–2021	780	15.0%	Model selection, thresholds, ablation, permutation importance
Test	2022–2024	1,359	15.0%	Final report only, after the model is frozen

Table 9. *Chronological split.*

- **Why chronological, not random:** a random split would let the model learn from 2023 to predict 2018, and form and standings features are strongly autocorrelated. A time split mimics real use: learn from the past, predict the future.
- **Why this cut:** validation (2020–2021) sits directly before the test period. The test seasons (2022–2024) are a **new regulation era**, so they are an honest test of generalisation.
- **Full history for training:** following the teaching-team clarification, all 1950–2019 races are used. Older data comes from a different sport, so we tested this choice (below), but did not drop any seasons.

Training starts in	2016–2017	2018–2019	2020–2021	Mean
1950	0.7256	0.6906	0.7139	0.7101
1960	0.7253	0.6914	0.7142	0.7103
1970	0.7189	0.6909	0.7130	0.7076
1980	0.7233	0.6893	0.7154	0.7093
1990	0.7179	0.6880	0.7148	0.7069
2000	0.7184	0.6857	0.7194	0.7078
2003	0.7138	0.6850	0.7195	0.7061
2005	0.7161	0.6857	0.7187	0.7069
2010	0.7158	0.6870	0.7143	0.7057
2014	0.7095	0.6940	0.7099	0.7045

Table 10. *Walk-forward diagnostic: validation PR-AUC of a logistic regression trained from different starting years, evaluated on three earlier validation periods.*

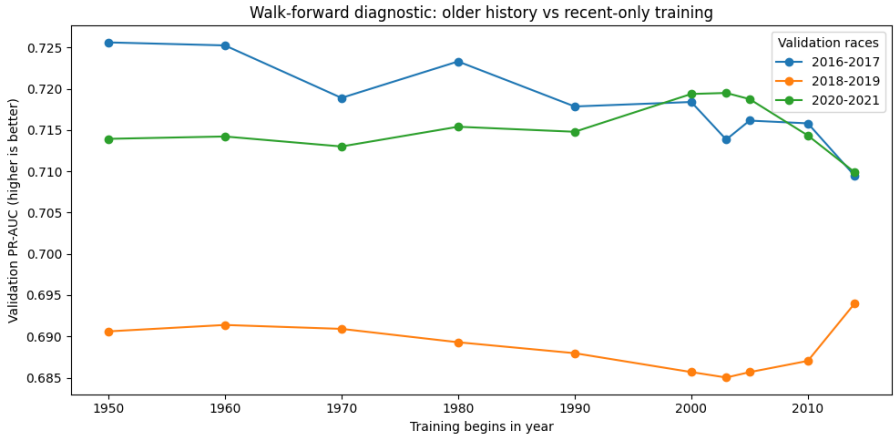


Figure 16. *No training start year is consistently best: the curves are flat to within roughly 0.01–0.016, and 2018–2019 is hard for every window. Keeping the full history (1950) costs nothing.*

4.1 Preprocessing

Fitted on the **training rows only** and then applied unchanged to validation and test: (1) a missing qualifying position falls back to the grid position, every other missing value takes the **training median**; (2) StandardScaler gives each feature mean 0 and standard deviation 1 (needed for logistic regression and the networks because, for example, drv_starts_before ranges up to 350 while points shares lie in 0–1; random forests do not need it). Effect: missing cells in the training features fall from **45,716 to 0**.

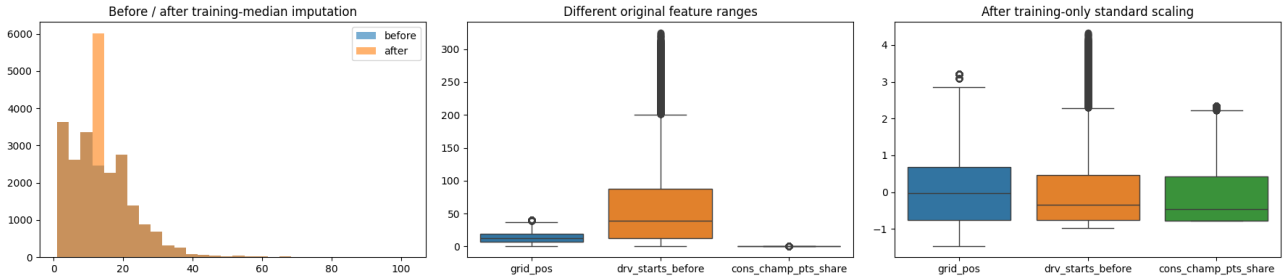


Figure 17. *Effect of preprocessing: median imputation (left), very different original feature ranges (middle), and the same features after scaling (right).*

Experiment (logistic regression, validation PR-AUC)	PR-AUC
1A impute, then scale (kept)	0.7139
1B scale, then impute	0.7085
2A add “was missing” flags before imputation	0.7157
2B add flags after imputation	0.7139

Table 11. *Two preprocessing-order experiments.*

Reading: the order of imputation and scaling changes validation PR-AUC by about 0.005, and adding missing-value flags by about 0.002. Both are small, so we keep the standard order (impute, then scale). All preprocessing decisions use validation data only.

5 Models and results

All models use the **same rows and the same 13 features**, so their scores are directly comparable. Thresholds are chosen on validation F1 only; hyperparameters and the final model are chosen on validation only. Six candidates are compared (five models plus a simple rule), which is more than the three attempts the brief requires.

Model	How it works	Main limitation
Rule: grid ≤ 3	Predicts “podium” if the car starts in the top three. The simplest reasonable reference.	Ignores car strength, form and everything behind P3.
Logistic regression (statistical baseline)	A linear model of the log-odds of a podium on scaled features; probabilities come from the sigmoid function. Easy to explain.	Cannot learn interactions (for example that pole matters more for the fastest team) unless hand-crafted.
Random forest (depth 8, and unlimited)	300 decision trees, each trained on a bootstrap sample; the forest averages their votes. Learns non-linear patterns without scaling.	The unlimited forest memorises the training data (train PR-AUC 1.000 vs validation 0.668). Hard to read as a whole.
Shallow neural network (Adam / SGD)	13 inputs → 32 ReLU units → 16 ReLU units → 1 sigmoid output; binary cross-entropy; batch 32; early stopping on validation loss (patience 10). Adam stopped after 32 epochs, SGD after 112.	Needs scaled inputs; on small tabular data it gains little over logistic regression; the hardest to explain; results vary slightly between runs.

Table 12. *The models, how they work, and their limitations.*

Primary metric: PR-AUC. Secondary metric: F1 (at the validation-tuned threshold). Only about 13–15% of driver-races are podiums, so accuracy is misleading: a model that always answers “no podium” would already score about 85%, and every model here scores 87–90%. ROC-AUC is also flattered by the many easy negatives. PR-AUC looks only at how well true podium finishers are ranked above the rest, which is the actual task. F1 shows how precision and recall balance at one concrete decision threshold. ROC-AUC, precision, recall and accuracy are reported as well.

5.1 Results on training, validation and unseen test data

Model	ROC-AUC	PR-AUC	F1	Precision	Recall	Accuracy
Rule: grid ≤ 3 (baseline)	0.890	0.493	0.541	0.541	0.541	0.885
Logistic regression	0.905	0.562	0.553	0.567	0.540	0.891
Random forest (depth 8)	0.928	0.657	0.605	0.622	0.590	0.904
Random forest (unlimited)	1.000	1.000	0.977	0.954	1.000	0.994
Neural network (Adam) – selected	**0.912**	**0.582**	**0.582**	**0.544**	**0.626**	**0.888**
Neural network (SGD)	0.911	0.583	0.578	0.543	0.618	0.888

Table 13. *Training set (1950–2019).*

Model	ROC-AUC	PR-AUC	F1	Precision	Recall	Accuracy
Rule: grid ≤ 3 (baseline)	0.909	0.639	0.692	0.692	0.692	0.908
Logistic regression	0.927	0.714	0.700	0.692	0.709	0.909
Random forest (depth 8)	0.929	0.708	0.706	0.703	0.709	0.912
Random forest (unlimited)	0.920	0.668	0.684	0.605	0.786	0.891
Neural network (Adam) – selected	**0.924**	**0.715**	**0.702**	**0.680**	**0.726**	**0.908**
Neural network (SGD)	0.926	0.712	0.707	0.674	0.744	0.908

Table 14. *Validation set (2020–2021). The network is selected here, by PR-AUC.*

Model	ROC-AUC	PR-AUC	F1	Precision	Recall	Accuracy
Rule: grid ≤ 3 (baseline)	0.914	0.647	0.649	0.650	0.647	0.895
Logistic regression	0.937	0.713	0.652	0.647	0.657	0.895
Random forest (depth 8)	0.942	0.714	0.671	0.653	0.691	0.898
Random forest (unlimited)	0.937	0.710	0.659	0.532	0.868	0.865
Neural network (Adam) – selected	**0.932**	**0.700**	**0.645**	**0.605**	**0.691**	**0.886**
Neural network (SGD)	0.935	0.710	0.661	0.610	0.721	0.889

Table 15. *Unseen test set (2022–2024), scored once after the model and threshold were frozen.*

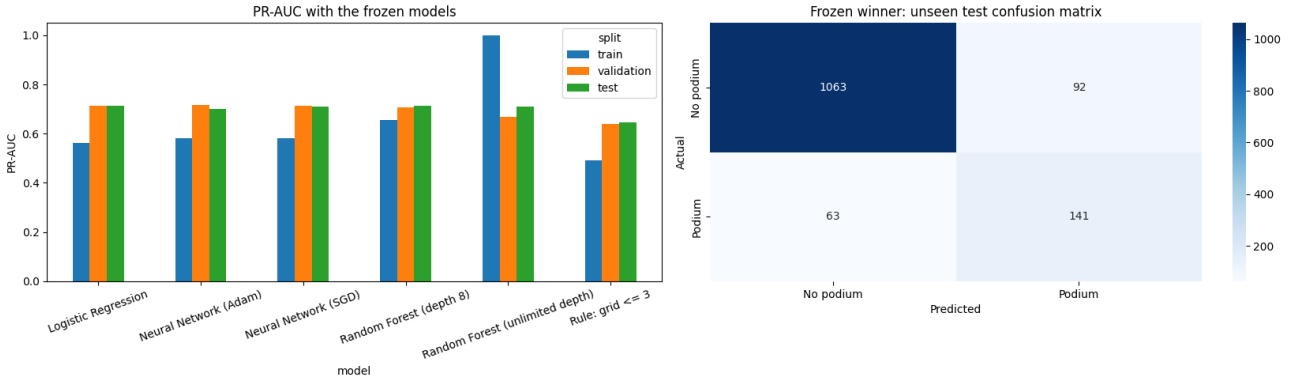


Figure 18. *Left: PR-AUC of each model on train, validation and test. Right: confusion matrix of the selected model on the test set (1,063 true negatives, 92 false positives, 63 missed podiums, 141 correct podiums).*

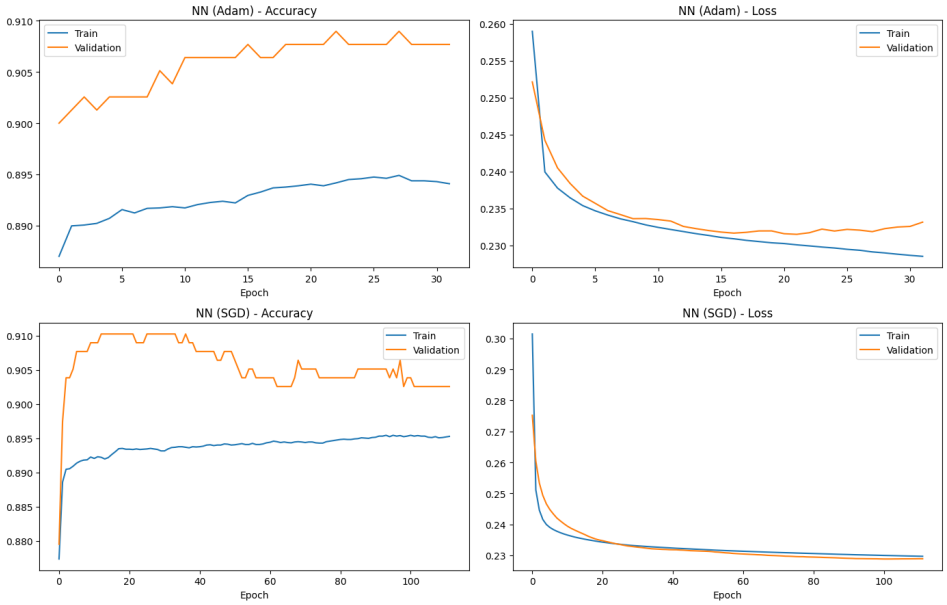


Figure 19. *Training and validation accuracy and loss per epoch for the two networks. Early stopping keeps the weights with the best validation loss.*

What the numbers honestly say. (1) Every learned model beats the grid ≤ 3 rule on the ranking metrics: test PR-AUC 0.700–0.714 vs 0.647 and test ROC-AUC 0.932–0.942 vs 0.914. (2) At a fixed threshold the advantage is small: test F1 is 0.645–0.671 for the models vs 0.649 for the rule, so the selected network is not better than the rule on F1. (3) The selected network (validation PR-AUC 0.715) is **not meaningfully better than the logistic regression baseline** (0.714); on test, logistic regression (0.713) and the depth-8 forest (0.714) are slightly above the network (0.700). We keep the validation-selected model because choosing by test score would turn the test set into a selection tool. The practical conclusion is that the **starting grid already carries most of the predictive signal**; the extra features add a modest improvement in ranking, and the model should not be presented as a large improvement over the rule.

6 Feature-group ablation

We remove one feature group at a time, retrain the **same selected model** (the Adam network) on the remaining features with the same seed, and compare validation PR-AUC with the all-feature model (0.7155). Test data is deliberately not used here, so the features are not tuned against the final exam.

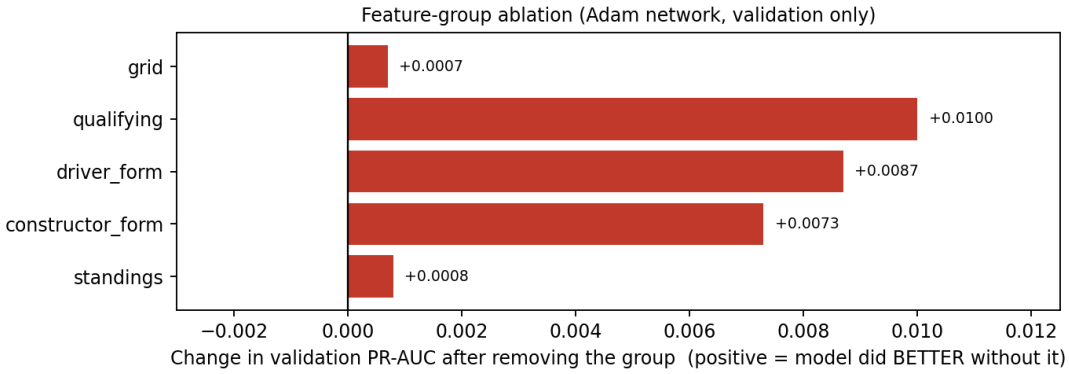


Figure 20. *Change in validation PR-AUC when a group is removed. Positive (red) means the model did better without the group.*

Removed group	Validation PR-AUC	Change vs all features
none (all features)	0.7155	–
grid	0.7162	+0.0007
qualifying	0.7255	+0.0100
driver_form	0.7242	+0.0087
constructor_form	0.7228	+0.0073
standings	0.7163	+0.0008

Table 16. *Leave-one-group-out ablation (validation only).*

How to read this. Removing any single group did *not* hurt validation PR-AUC (changes between +0.0007 and +0.0100), so this experiment cannot show that any group improves the network. Two reasons: the groups overlap strongly (grid and qualifying position correlate at 0.97; standings, team form and driver form all describe car strength), and a retrained network varies from run to run. A second full run of the same notebook in another environment gave changes between −0.0015 and +0.0106, with the sign flipping for the grid and standings groups, so differences below about 0.01 are noise. The features were therefore kept because of the pre-training evidence (Section 3.2) and the explanations in Section 7, not because the ablation proves each group helps. A natural next step is to repeat the ablation with a deterministic model such as logistic regression.

7 Explainability (XAI)

Explanations are not causal effects. A feature being important to the model only means the model relies on it to make predictions. It is not evidence that the feature causes podium finishes. Grid position, for example, is a proxy for car speed and qualifying form.

7.1 Global: permutation importance

We shuffle one feature at a time on the **validation** set and measure how much PR-AUC drops; the model is not retrained. A bigger drop means the model relies on that feature more.

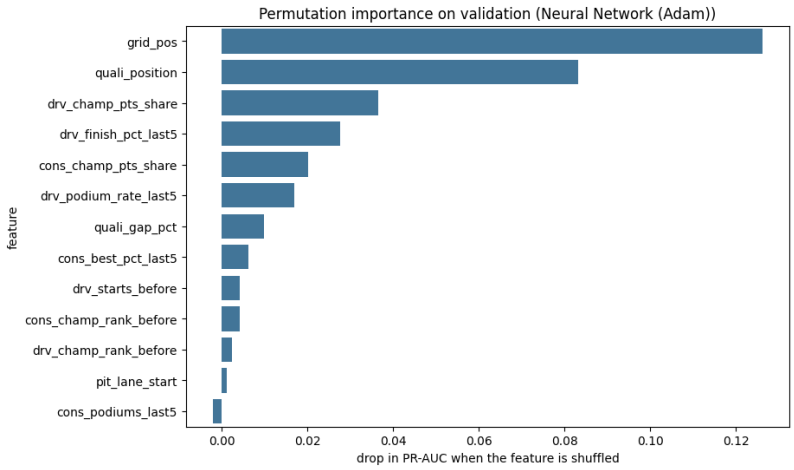


Figure 21. *Permutation importance of the Adam network on the validation set.*

Feature	Drop in validation PR-AUC	Std (repeats)
grid_pos	0.1261	0.0213
quali_position	0.0832	0.0213
drv_champ_pts_share	0.0365	0.0052
drv_finish_pct_last5	0.0277	0.0163
cons_champ_pts_share	0.0201	0.0148
drv_podium_rate_last5	0.0169	0.0050
quali_gap_pct	0.0099	0.0196
cons_best_pct_last5	0.0063	0.0055
drv_starts_before	0.0043	0.0198
cons_champ_rank_before	0.0042	0.0042
drv_champ_rank_before	0.0024	0.0044
pit_lane_start	0.0013	0.0025
cons_podiums_last5	-0.0019	0.0083

Table 17. *Permutation importance (mean over repeats). Values near or below zero mean no reliance.*

Reading: grid_pos (0.126) and quali_position (0.083) dominate, followed by the driver's points share (0.037), recent finishing percentile (0.028), the team's points share (0.020) and recent podium rate (0.017). cons_podiums_last5, pit_lane_start and the rank features contribute almost nothing. Because grid_pos and quali_position are 0.97 correlated, their importance is shared between them and should be read together.

7.2 Global: SHAP

SHAP splits each prediction into additive contributions per feature relative to the average prediction. For a neural network we use the model-agnostic Kernel SHAP, with a training background, on a small sample of 30 validation driver-races, a 30-row training background and 100 sampled coalitions per row. Kernel SHAP is slow, so these values are an **approximation** chosen for speed.

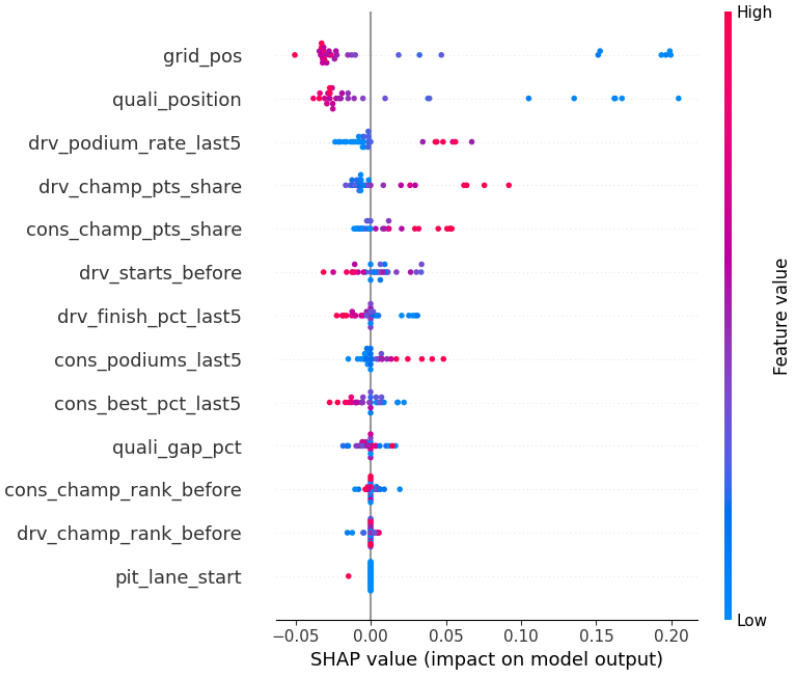


Figure 22. *SHAP beeswarm: one dot per driver-race; colour is the feature value (red high, blue low); position is the push toward (right) or away from (left) a podium.*

Reading: the same two features lead. Low grid_pos and quali_position values (blue, near the front) push strongly toward a podium, and a high driver points share, a high driver podium rate and a high team points share (red) push upward. This agrees with permutation importance and with the correlation evidence in Section 3.2, which raises our confidence that the model uses sensible information.

7.3 Local explanations for single predictions

Two unseen **test** predictions, explained after the model was frozen: the most confident podium call, and a hard case – a real podium from far down the grid.

- **Case A – 2024 Miami Grand Prix, Max Verstappen:** starts P1, predicted podium probability **0.802**, actual podium: yes.
- **Case B – 2024 São Paulo Grand Prix, Pierre Gasly:** starts P13, predicted probability **0.003**, actual podium: yes (an upset).

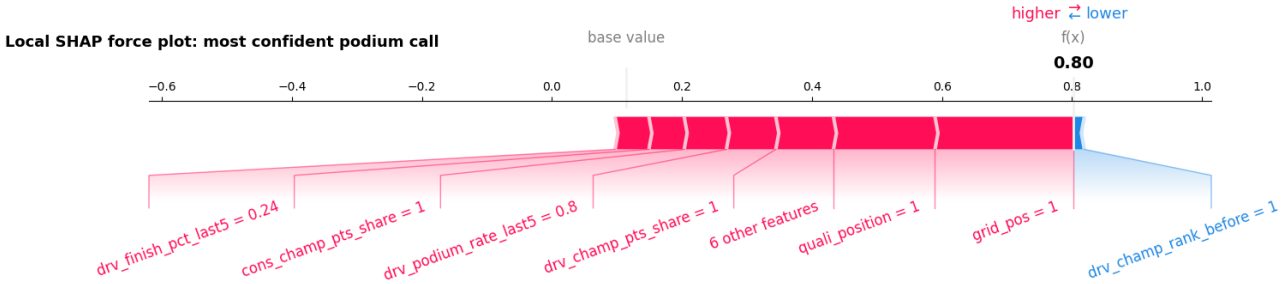


Figure 23. *Local SHAP force plot, case A. Pole position, front-row qualifying, a full share of the leader's points for driver and team, and a high recent podium rate push the prediction up; only a small blue term (drv_champ_rank_before) pushes it down. The six smallest contributions are summed as "6 other features".*

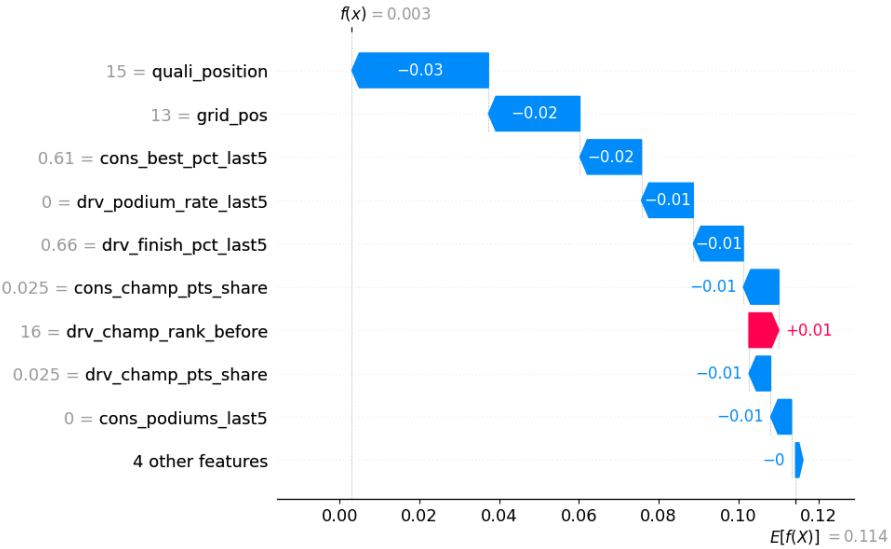


Figure 24. *Local SHAP waterfall plot, case B (base value 0.114 → prediction 0.003). Qualifying 15th, grid 13th, a weak recent team record, no recent podiums and a low points share all push the prediction down.*

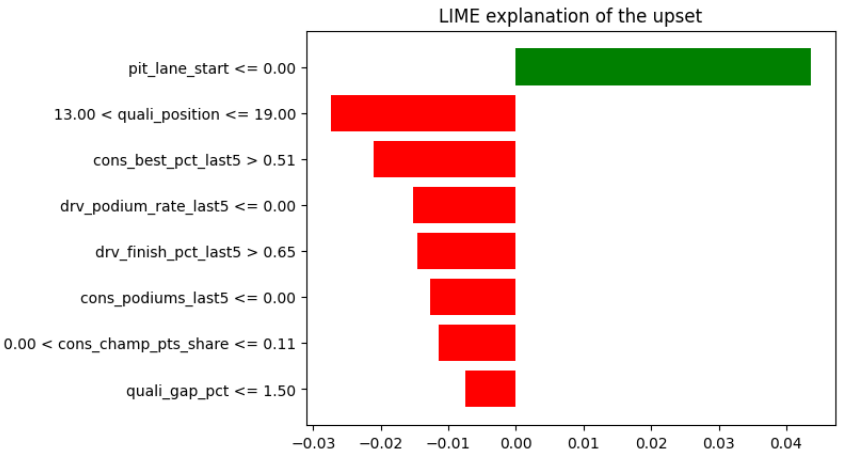


Figure 25. *LIME explanation of case B. LIME fits a simple local model around this one row; the conditions quali_position 13–19, cons_best_pct_last5 > 0.51 and no recent podiums push away from a podium.*

Reading: SHAP and LIME agree on why the model doubted case B: a poor qualifying slot and a weak recent record. The only pull toward a podium is "not a pit-lane start". The podium itself came from in-race

events (weather, safety cars, incidents ahead) that are not observable at the prediction cutoff. This is a limit of any pre-race model, not a bug. SHAP gives additive contributions against a background; LIME is a local linear surrogate on perturbed samples, so the two need not match exactly. In case A, a small negative term for drv_champ_rank_before = 1 is an example of correlated features splitting credit, not evidence that leading the championship hurts.

8 Inference function

predict_podium(record) takes a dictionary of pre-race features. Any feature may be left out. The function applies exactly the training preprocessing (qualifying falls back to grid, other gaps take the training median, then scaling), runs the frozen model, and returns the podium probability, the decision at the frozen threshold (0.34) and the list of features that had to be filled in.

Record	P(podium)	Decision	Features filled in
Complete record (2024 Hungarian GP, Lando Norris; actual podium: yes)	0.645	podium	none
Partially missing: Q1 gap and all four standings features removed	0.629	podium	5
Only grid information supplied	0.270	no podium	11

Table 18. *The inference function on a complete and on partially missing records.*

The function always returns a prediction because missing values are filled with training medians; the more is missing, the more the prediction falls back to “an average driver at that grid slot”, so it should be read as less informed, not equally reliable.

9 Known limitations

- **Qualifying-coverage drift.** No qualifying data before 1994 and complete only from 2003; only about a third of training rows have a qualifying position. Older rows rely on the grid and training medians.
- **Era sensitivity.** Rules, reliability, field sizes and points systems changed many times. We use points shares and finish percentiles and test training windows, but the 2022 regulation change still shifts relationships; differences between windows are descriptive and do not prove an era caused them.
- **Prediction population.** All entries are kept so that post-race status cannot decide who is in the data. grid = 0 mixes historical non-starters with real pit-lane starts (a proxy flag), and the Indianapolis 500 rows (1950–1960) are retained. Descriptive questions Q2 and Q3 use actual starters only.
- **Shared drives.** The lowest-resultId rule is outcome-blind but may not identify every driver’s original car; a few 1950s labels are ambiguous.
- **Constructor identity.** Renamed teams get new ids, so their form and standings history restarts at the rename.
- **Small evaluation sets.** Validation has 780 rows (about 117 podiums) and test has 1,359 rows (204 podiums). Differences of about 0.01 PR-AUC between models, runs or ablations are within noise.
- **Model value.** The selected network is not clearly better than logistic regression or than the grid ≤ 3 rule on F1; the grid carries most of the signal.
- **No in-race information by design.** Weather, safety cars and crashes cannot be known before the start, so upsets (such as case B) cannot be predicted.
- **Not causal.** All explanations describe the model’s reliance on features and observed associations, never causes.

10 Where each brief requirement is covered

Brief requirement	Notebook section	This report
EDA before and after cleaning, with plots	2, 4	1.1, 1.6
Sentinels (\N) and semantic typing (dates, lap times → seconds)	3.1	1.2
Primary / foreign keys; raw data immutable	3.3	1.3
One row per (raceId, driverId); duplicate rule justified	3.5	1.4, 1.5
Q1, Q2, Q3 with joins, plots, grain, missingness, cutoff	5	2
Feature choices with evidence; leakage-safe form features	6, 6.1, 6.2	3
Chronological split, justified	7	4
Preprocessing effect and ≥ 2 order experiments	8, 8.1	4.1
≥ 2 statistical models, a shallow network, baseline, ≥ 3 attempts	9, 9.1	5
ROC-AUC, PR-AUC, F1 on train / validation / test; metric choice justified	9.1	5, 5.1
Feature-group ablation	10	6
Permutation importance, SHAP, local SHAP, LIME; not-causal statement	11	7
Inference function on complete and partially missing records	12	8
Known limitations	13	9
Modeling table and result tables saved	14 (output folder acl_labstyle_results)	–

Table 19. *Traceability matrix.*

Reproducibility. The Kaggle notebook runs top to bottom with Save & Run All; random seeds are fixed (numpy and TensorFlow) and deterministic TensorFlow operations are enabled. The tables behind this report are written to the notebook’s output folder.